



Education

Indian Institute of Technology Madras (IIT M), India Expected: 2027	CGPA : 9.20/10
Pursuing 5th year in B.Tech + M.Tech, Aerospace Engineering	
Class XII, Central Board of Secondary Education, India 2022	CGPA : 9.86/10
Class X, Central Board of Secondary Education , India 2020	CGPA : 9.90/10

Publications & Conferences

- **Bhadra Sreelatha**, Rohit Radhakrishnan, R. I. Sujith. Tipping under rapid parameter variation in a thermoacoustic system. Manuscript under review in [Chaos: An Interdisciplinary Journal of Nonlinear Science](#). (Shock-induced tipping in a thermoacoustic system, [arXiv:2603.14237](#))
- **Bhadra Sreelatha**, Rohit Radhakrishnan, R. I. Sujith. Experimental investigation of shock-induced tipping in a thermoacoustic system: [79th Annual Meeting of the APS Division of Fluid Dynamics \(APS DFD\) 2026 \[Poster\]](#), [International Conference on Vibro-acoustics & Mechano-materials \(ICVAM\) 2026 \[Oral\]](#), [Conference on Nonlinear Systems & Dynamics \(CNSD\) 2026 \[Poster\]](#), [Inter Group Meeting \(IGM\) 2026 \[IIT Bombay, Oral, session chair\]](#)

Achievements

- Awarded **JASSO** scholarship by Ministry of Education, Culture, Sports, Science & Technology (MEXT), Japan
- Achieved **1st** place in Asia & **21st** worldwide for rocket flight mission at the Spaceport America Cup 2024
- Achieved **8th** place for technical design report & **23rd** worldwide in drone flight mission at the SUAS 2025

Research Experience

CoE- CTCS¹,

PI: Dr. R. I. Sujith, Institute Chair Professor, IIT M

Pattern formation across transition routes to instability in turbulent combustion (Nov '25 - Present)

- Extracting spatio-temporal turbulent flow dynamics using **PIV** data from a thermoacoustic rig for axial instabilities
- Statistically quantifying vortex structures during smooth & abrupt transitions from combustion noise to instability
- Identifying coherent patterns in simultaneous **Mie-scattering & chemiluminescence imaging** in distinct routes

Shock-induced tipping in a laminar thermoacoustic system- the Rijke tube (Nov '23 - Jan '25)

- Demonstrated **first experimental evidence** of shock-induced tipping using a prototypical thermo-fluid system
- Discovered a threshold in heating-grid temperature beyond which, system tips to instability, facilitating prediction

Experimentally explaining hysteresis during thermoacoustic instability in Rijke tube (Mar - Oct '25)

- Enhanced the Rijke-tube model to incorporate unsteady heat transfer & obtain temperature bifurcation diagram
- Experimentally captured the hysteresis dynamics underlying subcritical Hopf bifurcation to limit-cycle oscillations

Designed injector model to mitigate high frequency instability in rocket propulsion (June - July '24)

Liquid Propulsion Systems Centre, ISRO²

Guide: Mr. Aadithya Vijayakumar

- Modeled resonator-feed configurations for centrifugal & constant-rate pumps to determine most stable configuration
- Analyzed efficiency of Helmholtz & Quincke dampers (resonators) in rocket propellant feed systems using Python
- Diagnosed **critical flow stability** risks via AutoCAD-optimized injector-feedline admittance response analysis

Reproduction of low-gravity combustion using decompression environment (May - Aug '26)

Laboratory of Space Utilization, Hokkaido University, Japan

Guide: Asst. Prof. Yusuke Konno

- Predicted microgravity effects on flame shapes across $O_2\%$ by enhancing Roper laminar diffusion flame length model
- Established **flame shape scaling** with $O_2\%$ through microgravity-analog experiments using low-pressure surrogate

¹Center of Excellence for studying Critical Transitions in Complex Systems

²Indian Space Research Organization

Engineering Projects

Industrial Research Intern at Kinben Innovation Pvt. Ltd.

(May – July '25)

- Engineered EKF-based state estimation trajectory reconstruction of oil & gas pipelines in GPS-denied environments
- Modeled pipeline inspection gauge dynamics via CFD (OpenFOAM) to detect structural anomalies in the pipelines

Student rocketry team– Spaceport America (SA) Cup 2024

(May '23 – June '24)

Center for Innovation, IIT M

Guide: Prof. Satya Chakravarthy, IIT M

- Engineered India's first **SRAD hybrid rocket engine** with liquid nitrous oxide as oxidizer & paraffin as the fuel
- Simulated parachute dynamics in ANSYS Fluent & built a flight computer for recovery of rocket from 10,000 ft

Autonomous drone project, Student Unmanned Aerial Systems (SUAS) 2025

(May '24 – June '25)

Guidance, Navigation & Controls Lab, Website

Guide: Prof. Satadal Ghosh, IIT M

- Spearheaded a team of 13 to build a fully autonomous UAV using ROS framework equipped with computer vision
- Designed & optimized UAV frame through **structural stress analysis** in ANSYS, validating its structural integrity

Unmanned fixed-wing aircraft project, Design of UAVs

(Jan '25 – Jan '26)

Department of Aerospace Engineering, IIT M

Guide: Prof. HSN Murthy, IIT M

- Aerodynamically & structurally engineered a long-endurance UAV for maritime rescue missions in extreme weather
- Designed, manufactured & flight tested the fixed-wing aircraft fully in-house, from raw composites to final assembly

Course Projects

Combustion & Flow Diagnostics (Ongoing)

Guide: Prof. R. I Sujith

- Developing Schlieren-based diagnostics to extract burner flame geometry & dynamic response to flow perturbations

Acoustic Instabilities in Aerospace Propulsion

Guide: Prof. R. I Sujith

- Generated phase space, recurrence plots & complex network plot to characterize instability in turbulent combustor

Vapor Transport studies, Introduction to Multiphase Flows

Guide: Prof. Mahesh Panchagnula

- Applied Euler–Euler & Euler–Lagrange models to simulate vapor bubble transport across condensation boundaries

Turbulence -Flame interaction, Introduction to Turbulence

Guide: Prof. Sameen A

- Computed turbulent mixing energy in turbulence–flame interactions in a bluff-body stabilized combustor using LES

2D-Unsteady Solver in MATLAB, Computational Fluid Dynamics

Guide: Prof. Vagesh D. N

- Formulated a hybrid finite-volume solver for 2D unsteady convection–diffusion systems using Gauss–Seidel & TDMA

Relevant Coursework & Skills

Teaching Assistantship: Low speed Aerodynamics laboratory

Courses: Combustion & flow diagnostics, Computational Fluid Dynamics (CFD), Control Systems, Introduction to Multiphase-Flows, Introduction to Turbulence, Acoustic instabilities in aerospace propulsion, Propulsion for Aircraft & Rocket Systems, Structural dynamics & Aeroelasticity, Unsteady Aerodynamics

Skills: Programming Languages : Python, MATLAB, Octave, C++, | **CFD :** OpenFOAM, ANSYS Fluent

Modelling Software: Fusion-360, ANSYS , AutoCAD, XFOIL, XFLR5, MATLAB SIMULINK

Electronic Systems: Eagle, KiCad, Arduino, EasyEDA | **Practices & Management:** L^AT_EX, Github, MS Office

Extra-Curricular Activities

- SmartLock360, Product Design Club, Center For Innovation, IIT M:** Designed & developed a prototype of an anti-theft automatic cycle lock. Conducted market analysis to ascertain a target market & estimate cost
- Inter-Collegiate Dance Team, IIT M :** Secured top positions in multiple **national-level dance competitions**
- As an **Associate Manager at the Entrepreneurship-Cell**, coordinated & launched national-level events